Differential Equation

Mind map

Basics

- Ordinary / partial
- General: n constants
- Particular: $y(x_0) = y_0$

Order and degree

- Order: highest derivative
- Degree: after polynomial form
- $\sin y'$: degree undefined

Formation

- Differentiate n times
- Eliminate constants
- Count independent constants

Variable separable

- $\int f\,dx = \int g\,dy$
- $+\ln c$
- $t = ax + by + c$

Polar substitution

- $x\,dx + y\,dy = r\,dr$
- $x\,dy - y\,dx = r^2 d\theta$

Geometric curves

- $Y - y = m(X - x)$
- Initial ordinate $y - mx$
- Subtangent, subnormal

Integral identity

- Differentiate both sides
- $x = a$ gives $f(a)$

Homogeneous

- $y' = F\left(\frac{y}{x}\right)$
- $y = vx$
- Shift to (h, k)

Linear

- $y' + Py = Q$
- IF $= e^{\int P\,dx}$
- Linear in x

Bernoulli / substitution

- $z = y^{1-n}$
- $t = f(y)$

Rate problems

- $N = N_0 e^{kt}$
- Cooling, mixture

Inequalities

- $f' > kf \Rightarrow (e^{-kx} f)' > 0$

Orthogonal trajectory

- $y' \to -\frac{1}{y'}$
- Fixed angle: isogonal

Exact differential

- $d(xy),\ d\left(\frac{y}{x}\right)$
- $d\left(\tan^{-1}\frac{y}{x}\right)$
- Group terms